

Question:

The feedback loop in men which regulates the production of sex hormones includes all but which of the following?

Answer Choices:

- A. follicle-stimulating hormone
- B. oxytocin
- C. pituitary gland
- D. hypothalamus
- E. gonadotropin-releasing hormone
- F. luteinizing hormone
- G. testosterone

Reasoning + Answer:

<think>

The primary hormonal feedback loop regulating sex hormone production in men is centered around the hypothalamic-pituitary-gonadal axis. Stimulation of the hypothalamus by gonadotropin-releasing hormone (GnRH) sends a signal to the pituitary, which then releases follicle-stimulating hormone (FSH) and luteinizing hormone (LH). These gonadotropic hormones act primarily on the testes: FSH promotes spermatogenesis in Leydig cells and seminiferous epithelium; LH causes Leydig cells to synthesize and secrete testosterone. Testosterone then acts peripherally (e.g., mediating secondary sexual characteristics and maintaining libido) and centrally via hypothalamic negative feedback on GnRH secretion.

While oxytocin is a neurohormone involved in social behaviors and reproductive processes in men (e.g., sexual arousal, ejaculatory facilitation, uterine contractions), it is not part of the core regulatory feedback loop that controls testosterone production. Thus, among the options listed, oxytocin is the component outside this loop.

Correct answer: B. oxytocin

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<answer>

B. oxytocin

</answer>

